Can Vision-Language Models Evaluate CFD? A Benchmark for Physical Plausibility Assessment of Flow Field Visualizations

Younghwan Joo[a,b,*]

[a]Energy Efficiency Research Division, Korea Institute of Energy Research, 152 Gajeong-ro, Yuseong-gu, Daejeon 34129, Republic of Korea
[b]Energy Engineering, University of Science & Technology, 217 Gajeong-ro, Yuseong-gu, Daejeon 34129, Republic of Korea

Abstract

Validating computational fluid dynamics (CFD) simulation results remains a manual bottleneck in automated pipelines, requiring expert visual inspection of flow field images. Vision-language models (VLMs) could close this automation gap, yet their capability for physics-grounded assessment of CFD visualizations is unknown. We introduce CFD-VisQA, the first benchmark for this task, comprising 10 canonical thermal-fluid scenarios with 60 OpenFOAM cases, 258 flow field images, and 279 physics-grounded questions spanning 6 systematically generated error types. Using an API-isolated, blind evaluation protocol with base64-encoded images and three independent trials per model, we evaluate Claude Opus 4.6, GPT-5.4, Gemini 3.1 Pro, and a domain expert on 30 common items under two conditions: setup-conditioned (with problem description) and image-only. The results reveal a striking rank reversal: with setup text, Claude leads at 88.9% (zero false negatives across all trials), followed by the expert at 73.8%, GPT-5.4 at 67.8%, and Gemini at 57.8%; without setup text, the expert leads at 66.7% while Claude drops to 33.3%, the lowest among all evaluators. Claude's accuracy drop of 55.6 percentage points—the largest observed—confirms that its strength derives from systematic cross-referencing of stated parameters against visual features, a capability that collapses without textual context. The expert's stability (-7.1 pp) reflects gestalt assessment grounded in physical intuition.

*Corresponding author
Email address: yhjoo@kier.re.kr (Younghwan Joo)

We further demonstrate that subagent-based evaluation inflates VLM accuracy from 88.9% to 99.6% due to potential filesystem contamination, establishing API isolation as a methodological requirement for trustworthy VLM benchmarking. The benchmark dataset and evaluation pipeline are publicly available.

Keywords: vision-language model, computational fluid dynamics, benchmark, flow visualization, physical plausibility, automated validation

---

1. Introduction

Computational fluid dynamics (CFD) is increasingly embedded in automated design and analysis pipelines, from building ventilation assessment [1, 2] to industrial heat exchanger optimization. Recent work has demonstrated that vision-language models (VLMs) can extract geometric information from architectural images and automatically generate CFD simulation domains [3], while LLM agents can configure OpenFOAM case files from natural language descriptions [4, 5]. As these automated pipelines mature, a critical bottleneck persists: validating whether the resulting flow fields are physically reasonable.

In traditional CFD workflows, an experienced engineer inspects velocity contours, temperature fields, and streamline plots to confirm proper boundary layer development, expected recirculation patterns, and absence of numerical artifacts. This expert visual inspection cannot scale with the throughput of automated pipelines. If a VLM could perform this validation step—assessing whether a CFD flow field visualization is physically plausible—the automation loop would close entirely.

Yet it remains unknown whether current VLMs possess the domain-specific visual reasoning required for this task. Broad-scope benchmarks reveal that VLMs struggle with scientific imagery: PhysBench [6] reports that the best model achieves only 49.5% versus 95.9% human accuracy on physical world understanding, while MicroVQA [7] finds peak accuracy of 53% on microscopy images. In the engineering domain, DesignQA [8] and Picard et al. [9] demonstrate that VLMs falter on engineering documentation and design tasks. On visualization literacy, Pandey and Ottley [10] show that VLM accuracy drops from 76–96% on simple line charts to 18–61% on multi-encoding visualizations—the category to which CFD colormapped fields belong. The closest domain-specific work, ClimateIQA [11], finds that

VLMs "struggle with precise color identification and spatial localization" on meteorological heatmaps, a finding directly transferable to CFD contour plots.

A concurrent study by Ezemba et al. [12] provides the first direct evidence: evaluating seven VLMs on Ansys simulation images, they report that image-based accuracy (~53%) barely exceeds random chance for binary true/false questions, while text-based accuracy reaches 69.2%. However, their benchmark tests *comprehension* of correct simulations—whether VLMs can read and interpret simulation output—rather than *plausibility judgment*: whether VLMs can detect that a simulation result is physically wrong.

This distinction matters. An automated CFD pipeline does not need a VLM to describe what a flow field shows; it needs a VLM to flag when the result violates physical laws. Detecting that an inlet and outlet have been swapped, that buoyancy acts in the wrong direction, or that the solution has not converged requires reasoning about the consistency between stated boundary conditions and observed flow patterns—a fundamentally different task from visual comprehension.

This paper introduces **CFD-VisQA**, the first benchmark designed to evaluate VLM capabilities in assessing physical plausibility of CFD flow field visualizations. We evaluate three frontier VLMs and a domain expert under two information conditions—setup-conditioned (problem description provided) and image-only—revealing a striking rank reversal that exposes the mechanism underlying VLM performance on this task.

Our contributions are:

1. **CFD-VisQA benchmark dataset**: 60 OpenFOAM CFD cases across 10 canonical thermal-fluid scenarios, producing 258 flow field images with 279 physics-grounded questions and expert annotations, with both correct solutions and 6 types of intentionally generated errors at varying severity levels.
2. **API-isolated evaluation protocol**: We demonstrate that subagent-based VLM evaluation inflates accuracy (99.6% vs. 88.9%) due to potential filesystem contamination, establishing base64-encoded API calls as a methodological requirement for trustworthy VLM benchmarking.
3. **Rank reversal between evaluation conditions**: With setup text, Claude leads at 88.9% and the expert follows at 73.8%; without setup text, the ranking inverts (Expert 66.7% vs. Claude 33.3%). This reveals that VLM performance derives from setup–image cross-referencing rather

than visual understanding of physics.

4. **Model-specific setup dependency**: Claude drops 55.6 pp without setup text, GPT-5.4 drops 18.9 pp, while the expert drops only 7.1 pp. The magnitude of setup dependency is itself a diagnostic of each evaluator's assessment strategy, with implications for how VLMs should be deployed in CFD validation workflows.

2. Related Work

2.1. VLM Benchmarks for Scientific and Engineering Imagery

The rapid proliferation of VLMs has generated a parallel benchmarking effort, but CFD simulation output remains absent from all existing evaluation suites. MMMU [13] assembles 11,500 college-level questions across 30 disciplines; engineering and science subjects yield the lowest model accuracy, and the benchmark contains no flow field imagery. SciFIBench [14] evaluates 28 models on 2,000 scientific figure interpretation tasks from arXiv, concluding that VLM capabilities in scientific figure understanding are "poorly characterized." PhysBench [6] tests 75 VLMs on 10,002 physical-world understanding entries, but its fluid dynamics category covers everyday scenarios (water pouring, oil spreading) rather than simulation output. MathVista [15] extends evaluation to mathematical and scientific plots, while CharXiv [16] targets chart understanding from arXiv papers; both remain within the data-visualization domain.

Domain-specific benchmarks have begun to emerge for individual scientific fields. ClimateIQA [11] constructs 26,280 meteorological heatmap images with 762,120 instruction samples, finding that frontier VLMs struggle with colormap interpretation and spatial localization—challenges shared by CFD contour plots. In chemistry and materials science, MaCBench [17] benchmarks VLMs on domain-specific image reasoning, published in Nature Computational Science. For engineering, DesignQA [8] evaluates VLMs on CAD images and technical drawings, and Picard et al. [9] assess VLM performance across the engineering design pipeline. MicroVQA [7] targets microscopy at CVPR 2025, reporting peak accuracy of only 53%. SPIQA [18] and ArXivCap [19] address question answering on scientific paper figures more broadly.

Despite this breadth, no benchmark targets CFD simulation output images—velocity contours, temperature fields, streamline plots, or pressure distributions—nor the task of assessing whether such outputs are physically plausible.

2.2. AI for CFD: From Automation to Validation

The intersection of AI and CFD has been explored primarily through simulation setup automation and surrogate modeling, not result validation. CFDLLMBench [20] evaluates LLMs on CFD knowledge (92% on conceptual MCQs) and OpenFOAM case generation (25–34% success rate), but is entirely text-based. OpenFOAMGPT [4] implements a retrieval-augmented LLM agent for OpenFOAM configuration, while Dong et al. [5] fine-tune Qwen2.5 on 28,716 natural-language-to-OpenFOAM pairs, achieving 88.7% case generation accuracy. Wang et al. [21] provide a systematic LLM evaluation on CFD problem-solving, and AutoTurb [22] combines LLM-generated expressions with evolutionary search to discover turbulence model closures. These tools can generate CFD cases but cannot inspect whether the resulting flow fields are physically reasonable.

The extension to multimodal AI for CFD remains nascent. The most directly relevant work is a 2026 study [23] that fine-tunes Qwen2.5-VL on indoor CFD mixed-reality images, improving accuracy from below 30% to above 60%; however, the study is limited to a single architecture, a small dataset, mixed-reality overlays rather than standard post-processing output, and comprehension rather than error detection. DrivaerNet++ [24] contributes a large-scale multimodal automotive dataset with CFD simulations at NeurIPS 2025, but does not evaluate VLM understanding of the flow fields. In the building simulation domain, Dai et al. [25] explore GPT-4o for HVAC system design, and Alanis Ruiz et al. [26] employ generative models for indoor pollutant dispersion prediction.

Kashefi [27] provides a complementary negative result: generative AI systems (Midjourney, DALL-E 3, Gemini) produce physically incorrect fluid mechanics imagery when prompted with classical phenomena, attributed to the scarcity of CFD imagery in training data. This finding motivates the need for a public benchmark of authentic CFD visualizations with ground truth annotations.

2.3. VLM Physical Reasoning

A growing body of work investigates whether VLMs possess genuine physical understanding. Schulze Buschoff et al. [28], in Nature Machine Intelligence, find that VLM error patterns on intuitive physics tasks diverge substantially from human cognition. GRASP [29] reports that VLMs perform at or below 50% on Unity simulation-based physical reasoning. Ghaffari and Krishnaswamy [30] catalog specific failure modes in multimodal physical dynamics reasoning.

These results establish that current VLMs possess limited intuitive physics capabilities. However, domain-specific prompting can improve performance: providing structured context enables text–image cross-referencing that compensates for weak visual reasoning. CFD-VisQA's dual-protocol design (setup-conditioned versus image-only) directly tests this hypothesis, and our results confirm it—Claude achieves 88.9% with setup text but drops to 33.3% without, while the domain expert remains stable (-7.1 pp).

2.4. Flow Visualization and Deep Learning

Before VLMs, deep learning for flow visualization focused on task-specific feature detection. Liu et al. [31] survey vortex identification (VortexNet), shock wave detection, and flow regime classification from simulation images. These methods produce categorical outputs without natural language understanding or explanatory capability. Banerjee et al. [32] review over 250 papers on physics-informed computer vision, covering turbulent flow super-resolution and PINN-based reconstruction, but this literature predates the VLM era and lacks language-based reasoning.

2.5. Concurrent Work

Ezemba et al. [12] present the most directly comparable study, evaluating seven VLMs on 12 Ansys simulations (6 structural, 6 fluid) with 40 true/false questions. Their finding that image-based accuracy (∼53%) approaches random chance independently confirms that simulation imagery poses a distinct challenge for VLMs. However, their benchmark tests comprehension of correct simulations, not error detection; uses commercial software (Ansys) limiting reproducibility; omits expert comparison; and does not employ blind or isolation protocols. CFD-VisQA advances beyond this work by defining plausibility assessment as the evaluation task, systematically generating six error types, including a domain expert for calibration, and establishing an API-isolated protocol that addresses contamination risks (Section 3).

2.6. Research Gap

No existing benchmark evaluates VLM capability to assess physical plausibility of CFD simulation output images. Scientific VLM benchmarks cover textbook diagrams, data charts, meteorological heatmaps, microscopy, and engineering drawings—but not flow field visualizations. AI-for-CFD tools automate simulation setup and surrogate prediction—but cannot inspect results. Physical reasoning benchmarks test intuitive physics on everyday scenarios—but not quantitative reasoning about conservation laws, boundary conditions, and numerical artifacts in simulation output. CFD-VisQA fills this gap with the first dedicated benchmark combining physics-grounded plausibility questions, systematically generated errors, expert calibration, and an API-isolated evaluation protocol.

3. Benchmark Design

3.1. Design Philosophy

CFD-VisQA is built on three principles. First, the benchmark must measure a *spectrum* of visual understanding, from low-level colorbar reading to physics-grounded plausibility judgment, because a VLM that reads a maximum velocity value correctly may still fail to detect a reversed buoyancy plume. Second, the dataset must contain both *correct* flow fields and *intentionally erroneous* ones produced by controlled perturbations of the simulation setup; without ground-truth errors whose root causes are known, one cannot distinguish genuine physical reasoning from pattern matching. Third, the benchmark must be *fully reproducible*: every flow field is generated with OpenFOAM using published case files and scripts, and every visualization is rendered programmatically with fixed colormaps and resolutions.

3.2. Flow Scenarios

The benchmark comprises 10 canonical thermal-fluid scenarios (Table 1), selected to span a representative range of physics: natural, forced, and mixed convection; laminar and turbulent regimes; separation, recirculation, and convective instability. Reynolds numbers range from $\mathcal{O}(10^2)$ to $\mathcal{O}(10^4)$, and Rayleigh numbers from $\mathcal{O}(10^3)$ to 3×10^7, covering the transition from steady laminar to unsteady turbulent behavior. All simulations are two-dimensional and steady-state (or time-averaged for inherently unsteady cases).

Solver selection follows a simple rule: `simpleFoam` is used for isothermal incompressible flows, with the k–ω SST turbulence model activated when Re > 2000; `buoyantBoussinesqSimpleFoam` is used for all buoyancy-driven flows under the Boussinesq approximation. This two-solver approach avoids unnecessary complexity while faithfully representing the physics of each scenario.

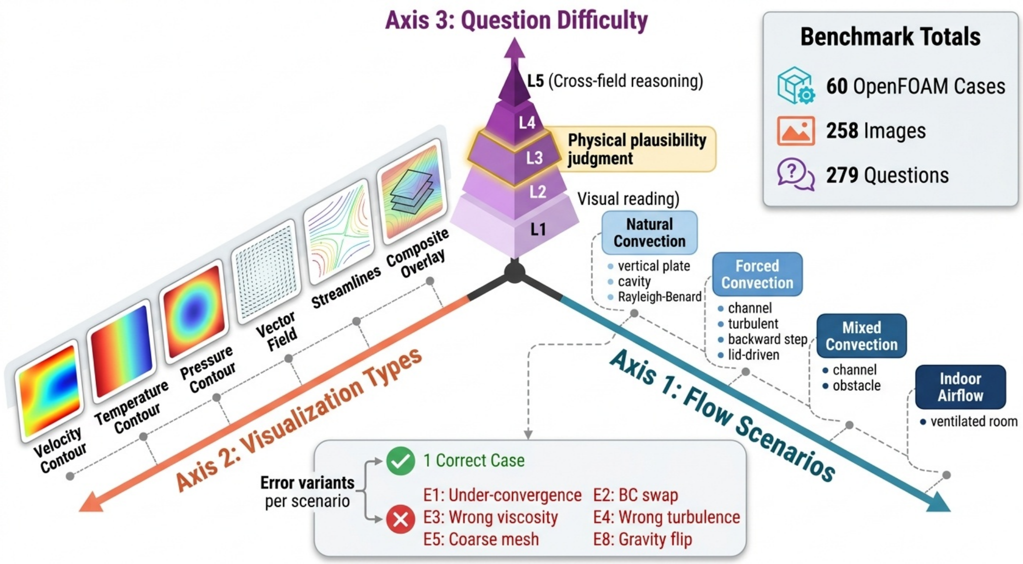

Figure 1: Three-axis structure of the CFD-VisQA benchmark: 10 flow scenarios (vertical), 6 visualization types (horizontal), and 5 question difficulty levels (depth). Each scenario includes one correct case and five erroneous variants, yielding 60 cases total.

Table 1: Ten canonical flow scenarios in CFD-VisQA. Each scenario produces 6 cases (1 correct + 5 error variants), yielding 60 cases total.

ID	Scenario	Physics	Solver	Cases
S1	Heated vertical plate	Natural conv.	buoyantBoussinesq[a]	6
S2	Channel flow (laminar)	Forced conv.	simpleFoam	6
S3	Channel flow (turbulent)	Forced conv.	simpleFoam + k-ω SST	6
S4	Backward-facing step	Separation	simpleFoam	6
S5	Lid-driven cavity	Recirculation	simpleFoam	6
S6	Differentially heated cavity	Natural conv.	buoyantBoussinesq[a]	6
S7	Mixed convection channel	Mixed conv.	buoyantBoussinesq[a]	6
S8	Heated obstacle in channel	Wake + thermal	buoyantBoussinesq[a]	6
S9	Rayleigh–Bénard conv.	Conv. cells	buoyantBoussinesq[a]	6
S10	Ventilated room	Indoor airflow	simpleFoam + k-ω SST	6
	Total			**60**

[a]`buoyantBoussinesqSimpleFoam`

The scenarios are grouped into three physical regimes. *Natural convection* (S1, S6, S9) is governed by the Rayleigh number, with $\mathrm{Ra} = 10^3$–3×10^7 spanning from steady laminar plumes to multi-cell Rayleigh–Bénard convection. *Forced convection* (S2–S5, S10) is governed by the Reynolds number, with $\mathrm{Re} = 100$–10^4 covering laminar developing flow through fully turbulent separation. *Mixed convection* (S7, S8) involves both buoyancy and inertial forcing, with the Richardson number $\mathrm{Ri} = \mathrm{Gr}/\mathrm{Re}^2$ determining the relative importance of each mechanism. This regime diversity ensures that VLM evaluation is tested across fundamentally different flow physics, not merely parametric variations of one scenario.

All meshes are structured or block-structured, generated with OpenFOAM's `blockMesh` utility. Baseline meshes use $\mathcal{O}(10^4)$ cells with grading near walls to resolve boundary layers; coarse-mesh error variants (E5) reduce this to $\mathcal{O}(10^3)$ cells. Convergence is assessed by residual reduction below 10^{-5} for all transport equations, except for under-convergence variants (E1) which are stopped at 30 iterations.

3.3. Error Taxonomy

Each of the 10 scenarios includes five erroneous variants, produced by controlled perturbation of one simulation parameter at a time. The six error types are:

- **E1: Under-convergence.** The simulation is stopped after only 30 iterations, well before residual convergence. The flow field retains initial-condition artifacts and intermediate numerical transients, producing visually noisy or partially developed patterns.

- **E2: Boundary condition swap.** Inlet and outlet, or hot and cold wall conditions, are exchanged. This produces mirror-image or reversed flows that may appear internally self-consistent but contradict the stated boundary conditions.

- **E3: Wrong viscosity.** The kinematic viscosity is perturbed by a factor of 10–100×, changing the effective Reynolds or Rayleigh number. The resulting flow field is physically self-consistent at the *wrong* non-dimensional parameters, making this error visually subtle.

- **E4: Wrong turbulence model.** A laminar solver is applied at turbulent Reynolds numbers, or a turbulence model is activated for a laminar flow. This alters boundary layer structure, separation behavior, and recirculation zone size.

- **E5: Coarse mesh.** The cell count is reduced by approximately $10\times$, introducing discretization artifacts: blocky contours, missing corner vortices, and staircase-like boundary representations.
- **E8: Gravity/direction reversal.** The gravity vector or lid driving direction is flipped, producing mirrored buoyancy plumes or reversed recirculation patterns.

A key design rationale governs this taxonomy: the errors span a spectrum from *visually obvious* (E1 under-convergence with noisy residuals; E5 coarse mesh with blocky contours) to *visually subtle but setup-inconsistent* (E2 boundary condition swap producing a plausible but reversed flow; E3 wrong viscosity yielding a self-consistent field at incorrect parameters). This spectrum is essential for discriminating between gestalt visual assessment, which catches gross artifacts, and systematic parameter–image cross-referencing, which catches subtle setup inconsistencies.

3.4. Visualization Standardization

All flow field images are generated through a deterministic rendering pipeline to eliminate visualization variability as a confounding factor. Two-dimensional contour plots are produced with matplotlib, while three-dimensional VTK data (where applicable) are rendered with PyVista. The pipeline enforces a fixed colormap convention: `jet` for velocity magnitude fields and `coolwarm` for temperature fields. All images are rendered at a fixed resolution of 1920×1080 pixels with standardized axis labels including physical units (m, $\mathrm{m\,s^{-1}}$, K) and a colorbar with min/max annotations on every image.

Six visualization types are generated for each case: velocity magnitude contour, velocity vector field, streamlines, pressure contour, temperature contour, and a composite overlay combining velocity vectors with temperature contours. All rendering scripts are included in the published repository, enabling exact reproduction of every image in the benchmark.

3.5. Question Hierarchy

The 279 questions in CFD-VisQA are organized into five difficulty levels:

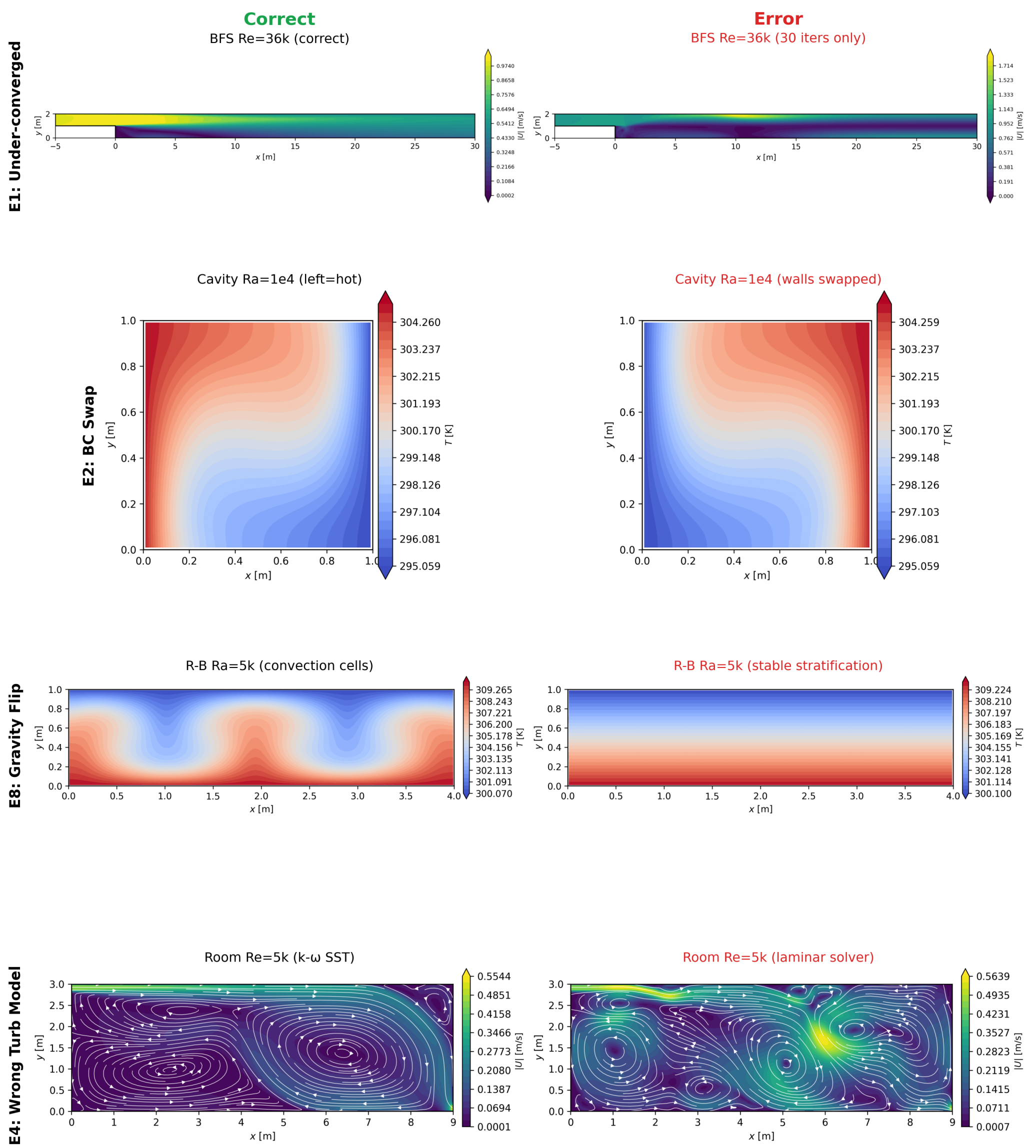

Figure 2: Examples of the six error types applied to the lid-driven cavity scenario (S5). Each perturbation produces a distinct flow field artifact, ranging from visually obvious (E1, E5) to visually subtle (E2, E3).

- **L1 (Visual reading):** "What is the maximum velocity in the domain?"—requires colorbar reading.
- **L2 (Pattern recognition):** "Is there a recirculation zone?"—requires identifying flow features from contour or streamline patterns.
- **L3 (Physics judgment):** "Is this flow field physically plausible given the stated conditions?"—requires domain knowledge to assess consistency between the image and the governing physics.
- **L4 (Quantitative reasoning):** "Is the boundary layer thickness consistent with the stated Reynolds number?"—requires dimensional analysis and order-of-magnitude estimation.
- **L5 (Cross-field reasoning):** "Are the velocity and temperature fields mutually consistent?"—requires multi-physics understanding across coupled transport equations.

The primary evaluation task in this paper uses **L3 binary plausibility assessment** (OK/Anomaly), as this level is the most practically relevant for automated CFD validation pipelines: it tests whether an evaluator can determine if a flow field is physically reasonable given the problem setup, which is precisely the judgment required to close the automation loop.

3.6. Expert Annotation Protocol

Ground-truth labels were produced by a single domain expert with over 10 years of experience in thermal-fluid CFD simulation. Annotations were performed via a blind evaluation protocol: flow field images were presented in a Google Drive document without case identifiers or error-type labels.

Two annotation conditions were applied. In the *setup-conditioned* evaluation, 80 items (covering all 10 scenarios and error types) were annotated; the expert received the full problem description—geometry, Reynolds/Rayleigh number, boundary conditions, solver type, and expected physical regime—alongside each flow field image. In the *image-only* evaluation, a 30-item subset was annotated with no textual context; the expert received only the rendered image.

In both conditions, the expert provided a binary verdict (OK or Anomaly) with free-text reasoning justifying the assessment. We explicitly acknowledge a limitation: this protocol relies on a single expert annotator. Multi-expert evaluation with inter-annotator agreement analysis is planned as future work.

4. Evaluation Protocol

4.1. API-Isolated Design

Each flow field image is base64-encoded and transmitted via direct API calls to three frontier VLMs: Anthropic Claude Opus 4.6 (model ID `claude-opus-4-6`), OpenAI GPT-5.4 (model ID `gpt-5.4`), and Google Gemini 3.1 Pro (model ID `gemini-3.1-pro-preview`). The evaluation enforces strict isolation: each VLM receives only image bytes and prompt text through the provider's API, with zero filesystem access, no project context, and no shared state between evaluation items. Each of the 30 evaluation items constitutes an independent API call. Default API temperature settings are used for all models, preserving the non-deterministic response variation inherent in production deployments.

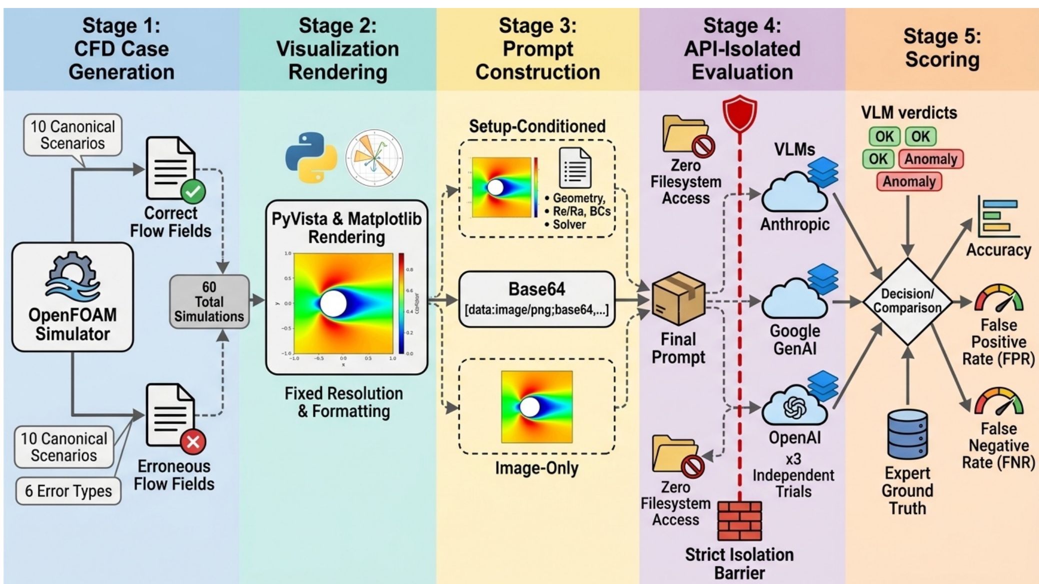

Figure 3: API-isolated evaluation protocol. Each flow field image is base64-encoded and sent as an independent API call. The VLM receives only image bytes and prompt text, with no filesystem access or shared state between items. This design eliminates the contamination pathway identified in Section 4.4.

4.2. Two Evaluation Conditions

Each evaluator—three VLMs and one human expert—is assessed under two conditions on the same 30-item subset.

Setup-conditioned. The flow field image is paired with a textual problem description specifying: domain geometry and dimensions, Reynolds or Rayleigh number, boundary conditions (wall temperatures, inlet velocity, or lid velocity), solver type and turbulence model, and expected physical regime. For example, a setup-conditioned prompt for a lid-driven cavity reads: *"2D lid-driven cavity, 1 m × 1 m domain, top wall moving at 1 m/s (left to right), Re = 100, laminar solver (simpleFoam), no-slip walls. Examine the velocity magnitude contour and assess whether the flow field is physically plausible."*

Image-only. The same image is presented with a generic prompt: *"Examine this CFD flow field visualization. Assess whether the flow field is physically plausible. Respond with OK if plausible, or Anomaly if you detect physically implausible features."* No scenario name, governing parameters, or boundary conditions are provided.

This paired design constitutes a deliberate ablation: by removing the setup text, we isolate the contribution of setup–image cross-referencing to evaluation accuracy. It is not two separate benchmarks but a controlled comparison within the same benchmark, enabling direct measurement of each evaluator's dependence on textual context.

4.3. Multi-Trial Protocol

To quantify response variability under default API temperature, each VLM undergoes 3 independent trials per condition, yielding a total of 3 models $\times$ 2 conditions $\times$ 3 trials = 18 evaluation runs. All runs evaluate the same 30 common items, selected as a stratified subset covering all 10 scenarios (3 items per scenario: 1 correct case and 2 error variants chosen to span the subtle–obvious spectrum). We report per-trial accuracy, mean accuracy, and per-trial false-positive and false-negative counts. The human expert completes 1 trial per condition, as single-trial human evaluation is standard practice and multi-trial human baselines are not meaningful given the absence of stochastic variation in expert judgment.

4.4. Contamination Discovery

During protocol development, we discovered that the evaluation *method* materially affects reported accuracy. Three isolation levels were tested for Claude:

1. **Subagent** (99.6% accuracy): The VLM runs as a subagent within an agent framework with filesystem access to the project directory containing ground-truth label files.

2. **CLI-isolated** (83.3%): The VLM runs via a separate CLI process with restricted tool access but a shared filesystem.
3. **API-isolated** (88.9%): The VLM receives only base64-encoded images via direct API calls with zero filesystem access.

The 10.7 pp inflation from API-isolated to subagent represents a contamination threat: the agent framework's filesystem access likely permitted reading ground-truth annotation files during evaluation. This finding establishes API isolation as a *methodological requirement* for trustworthy VLM benchmarking. All results reported in this paper use the API-isolated protocol exclusively.

5. Results

5.1. Overall Performance

Table 2 presents setup-conditioned accuracy across three independent trials for each VLM and one trial for the domain expert, evaluated on the common 30-item subset.

Table 2: Setup-conditioned evaluation accuracy (%) on the 30-item common subset. T1–T3 denote independent trials. FP and FN are trial-averaged counts.

Evaluator	T1	T2	T3	Mean	FP_{mean}	FN_{mean}
Claude Opus 4.6	90.0	90.0	86.7	88.9	3.3	0.0
Expert	73.8	—	—	73.8	2.0	19.0
GPT-5.4	56.7	70.0	76.7	67.8	0.7	9.0
Gemini 3.1 Pro	56.7	53.3	63.3	57.8	5.3	7.3

Claude achieves the highest mean accuracy (88.9%) with zero false negatives across all three trials: every intentionally generated error was detected, with only 3–4 correct cases over-flagged per trial. The domain expert ranks second at 73.8%, exhibiting low false-positive count (2) but a high false-negative count (19), indicating that nearly one-quarter of the errors went undetected. GPT-5.4 places third at 67.8% with notable trial-to-trial variance (56.7–76.7%), and accuracy that increases monotonically across trials (T1 < T2 < T3). Gemini ranks lowest at 57.8% with the highest false-positive rate (5.3 per trial), suggesting a tendency toward over-flagging that does not translate into error detection accuracy (Fig. 4).

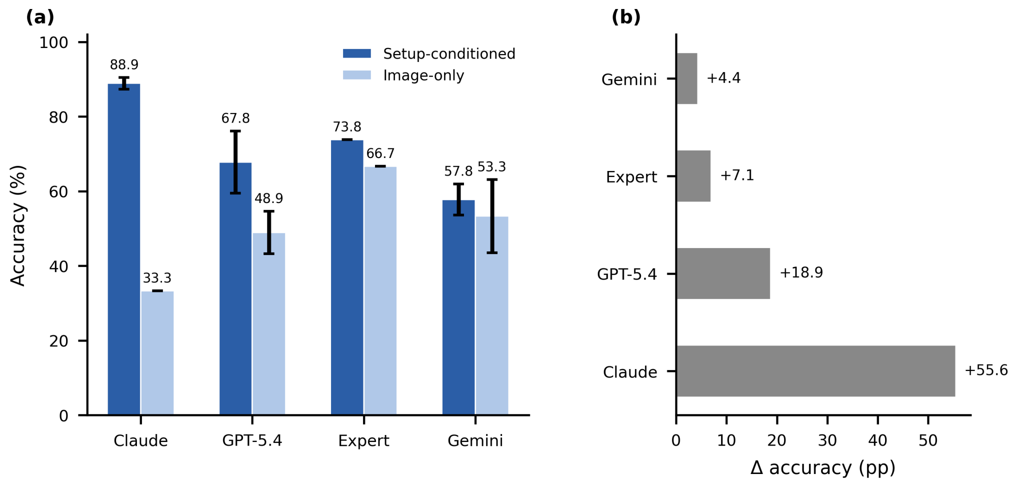

Figure 4: Setup-conditioned accuracy across evaluators. Error bars indicate the range across three independent trials (single trial for the expert). Claude achieves the highest mean accuracy with zero false negatives.

5.2. The Rank Reversal

Table 3 compares performance under setup-conditioned and image-only conditions, revealing a striking rank reversal (Fig. 5).

Table 3: Setup-conditioned vs. image-only accuracy (%) on the 30-item common subset. Δ denotes the change in mean accuracy when setup text is removed.

Evaluator	Setup mean	Image-only mean	Δ (pp)
Claude Opus 4.6	88.9	33.3	-55.6
Expert	73.8	66.7	-7.1
GPT-5.4	67.8	48.9	-18.9
Gemini 3.1 Pro	57.8	53.3	-4.4

With setup text, the ranking is Claude (88.9%) > Expert (73.8%) > GPT-5.4 (67.8%) > Gemini (57.8%). Without setup text, the ranking completely inverts: Expert (66.7%) > Gemini (53.3%) > GPT-5.4 (48.9%) > Claude (33.3%). Claude transitions from the best evaluator to the worst; the expert moves from second to first. Claude's -55.6 pp drop is the most extreme, while the expert's -7.1 pp drop makes it the most robust evaluator across conditions. Gemini is nearly unaffected (−4.4 pp), suggesting minimal reliance on setup–image cross-referencing. GPT-5.4 shows a moderate decline (−18.9 pp), intermediate between the extremes.

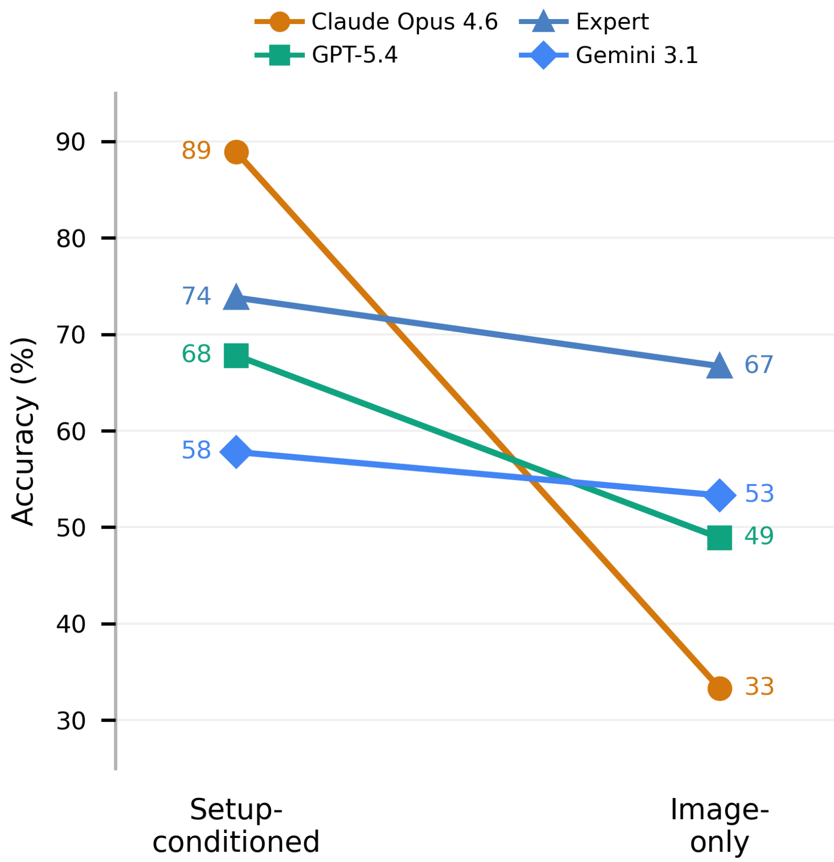

Figure 5: Rank reversal between evaluation conditions. With setup text (left), Claude leads; without setup text (right), the expert leads and Claude drops to the lowest accuracy among all evaluators.

5.3. Per-Error-Type Analysis

The false-positive and false-negative profiles differ systematically across evaluators (Fig. 6). Claude achieves FN = 0 across all error types when setup-conditioned, indicating 100% recall regardless of error subtlety. The expert's primary blind spots are concentrated on E2 (boundary condition swap) and

E3 (wrong viscosity)—precisely the error types where the flow field is qualitatively correct but inconsistent with the stated parameters. These are "subtle" errors: E2 produces a mirror-image flow that satisfies the governing equations under swapped boundary conditions, and E3 produces a self-consistent flow at the wrong Reynolds or Rayleigh number.

GPT-5.4 shows a distinct pattern: its false negatives span multiple error types (E1, E2, E3, E5, E8), with higher miss rates on under-convergence (E1) and wrong-viscosity (E3) cases. Gemini's errors are similarly distributed but with additional false positives—correct cases flagged as anomalous—particularly for visually complex flows with multi-vortex structures.

The BC swap illusion is particularly instructive (Fig. 7): a differentially heated cavity with swapped hot/cold walls produces a mirror-image temperature field with proper thermal stratification and well-developed boundary layer flow. The expert judges such cases as "typical natural convection" because the overall flow pattern is qualitatively correct. Claude, however, detects that the colorbar values contradict the stated boundary conditions—the left wall should be hot but the colorbar shows it as cold. This exemplifies the cross-referencing mechanism: Claude compares the stated wall temperature with the observed colorbar gradient, a comparison that requires both textual setup and image content.

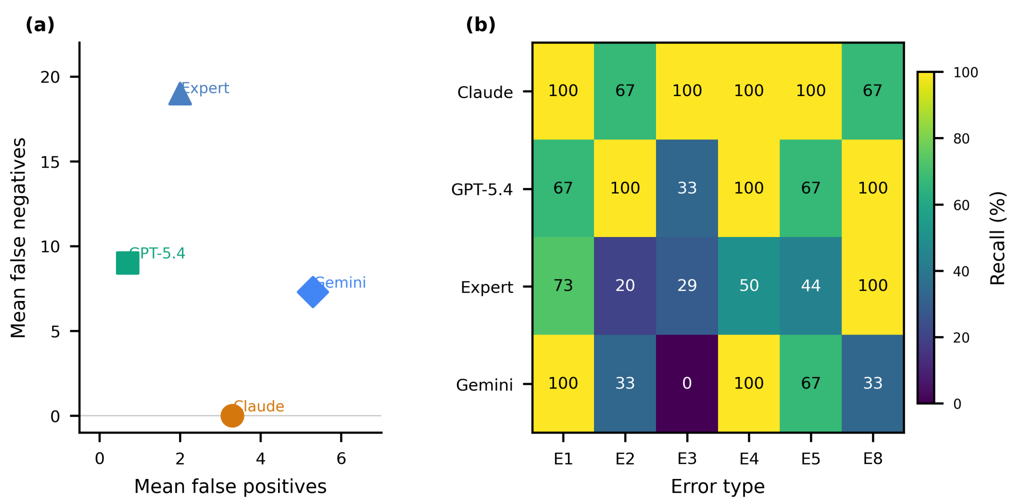

Figure 6: False-positive and false-negative profiles by evaluator. Claude achieves zero false negatives across all error types when setup-conditioned. The expert's false negatives concentrate on E2 and E3.

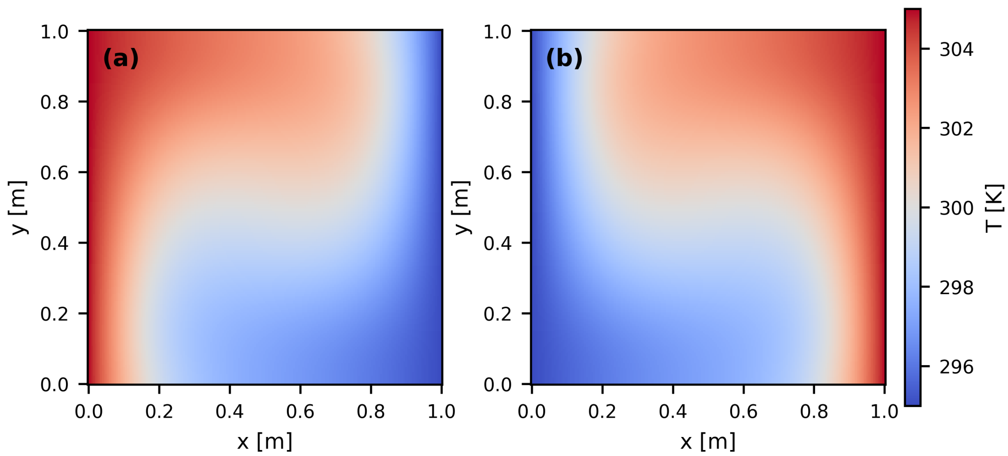

Figure 7: The BC swap illusion: a differentially heated cavity with swapped hot/cold walls produces a mirror-image flow that is internally self-consistent. The expert judges it as plausible; Claude detects the colorbar–boundary-condition mismatch.

5.4. Trial Consistency

Claude exhibits the highest trial consistency, with setup-conditioned accuracy ranging from 86.7% to 90.0% and image-only accuracy perfectly identical at 33.3% across all three trials (Fig. 8). GPT-5.4 shows notable trial variance (56.7–76.7% setup-conditioned), with accuracy increasing monotonically across trials (T1 $<$ T2 $<$ T3), possibly suggesting session-level learning effects within the API's default temperature sampling. Gemini exhibits the highest variance in both conditions (53.3–63.3% setup; 40.0–63.3% image-only).

5.5. Qualitative Case Studies

Three representative cases illustrate the distinct evaluation strategies. First, wrong-viscosity scenarios (Fig. 9) show that setup text enables detection of errors invisible to pure visual assessment: a channel flow with incorrect Reynolds number produces a plausible velocity profile that only contradicts the stated parameters, and a heated plate with incorrect Rayleigh number produces a smoother boundary layer than expected. Second, a coarse-mesh lid-driven cavity (Fig. 10) demonstrates that VLMs can detect discretization artifacts—blocky contours and missing corner vortices—that the expert overlooks. Third, a correct but complex heated obstacle wake (Fig. 11) triggers

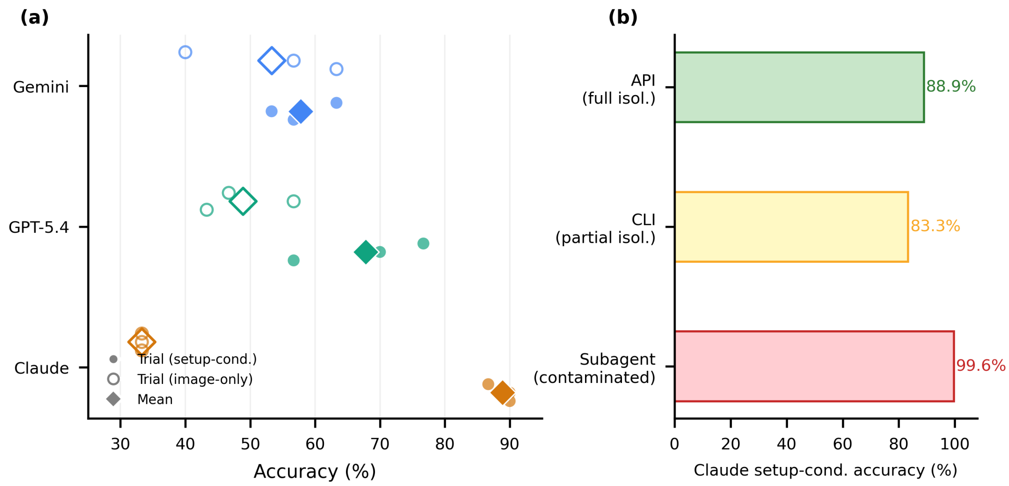

Figure 8: Per-trial accuracy for each evaluator under both conditions. Claude is the most consistent; GPT-5.4 shows monotonically increasing accuracy across setup-conditioned trials.

false positives from multiple evaluators, indicating that visual complexity can mimic error signatures even when the simulation is physically correct.

6. Discussion

6.1. Setup Text as the Critical Variable

The rank reversal between evaluation conditions is the central finding of this study. It reveals that VLMs do not understand CFD flow physics in the way a domain expert does; rather, their performance derives from a fundamentally different assessment strategy.

Claude and GPT-5.4 perform systematic cross-referencing: they compare stated parameters—Reynolds number, Rayleigh number, boundary temperatures, lid direction—against observable image features such as hydrodynamic development length, colorbar value ranges, vortex rotation direction, and thermal boundary layer thickness. When the textual anchor is removed, this cross-referencing strategy collapses entirely. Claude's -55.6 pp drop confirms that its 88.9% accuracy is almost exclusively attributable to setup–image comparison, not to any internalized understanding of fluid dynamics.

The expert, by contrast, employs gestalt assessment: evaluating whether the overall flow pattern "looks right" based on years of experience with

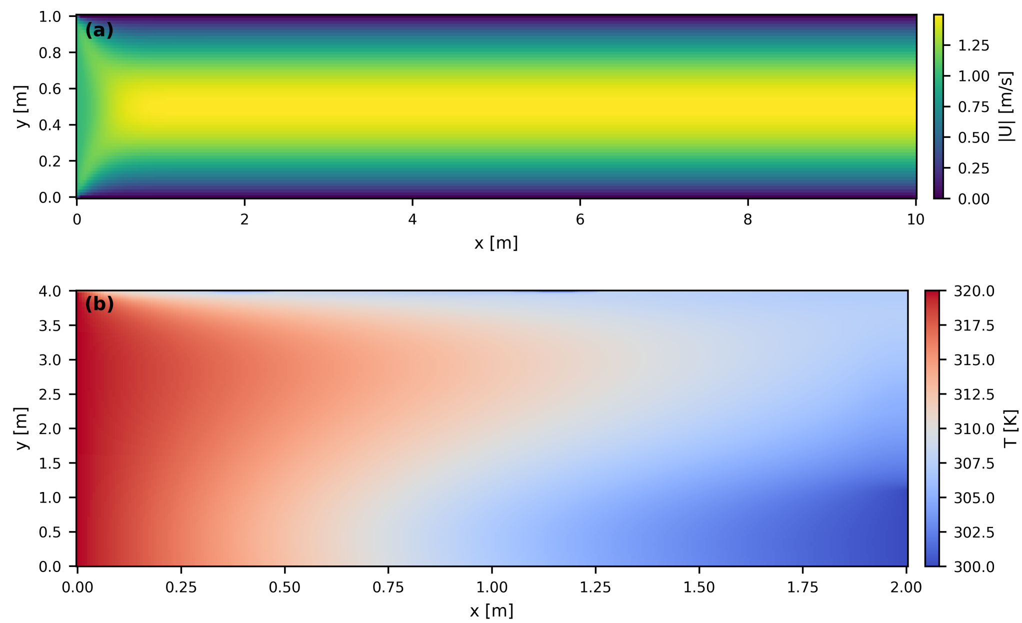

Figure 9: Wrong-viscosity cases where setup text enables error detection. Without stated parameters, the flow fields appear qualitatively plausible.

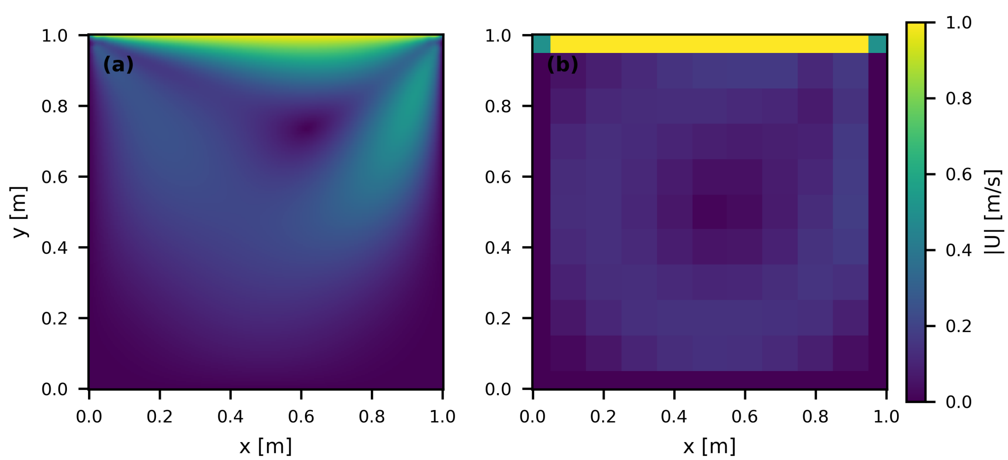

Figure 10: Coarse-mesh lid-driven cavity: VLMs detect discretization artifacts (blocky contours, missing corner vortex) that the expert overlooks.

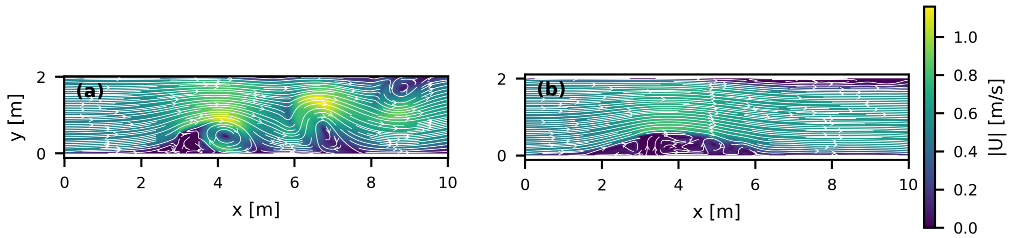

Figure 11: Correct heated obstacle wake triggering false positives: visual complexity mimics error signatures across multiple evaluators.

canonical thermal-fluid configurations. This strategy is robust to setup text removal (-7.1 pp) because it relies on internalized physical intuition rather than explicit parameter checking. The expert recognizes a properly developed boundary layer, a well-formed recirculation zone, or characteristic Rayleigh–Bénard convection cells without needing to verify them against stated dimensionless numbers.

However, this gestalt strategy is vulnerable to errors where the flow is qualitatively correct but quantitatively inconsistent with stated parameters. The expert's high false-negative count on E2 (boundary condition swap) and E3 (wrong viscosity) reflects precisely this limitation. A differentially heated cavity with swapped hot and cold walls produces a mirror-image flow that satisfies all qualitative expectations of natural convection. A channel flow with viscosity increased by a factor of 10 produces a smoother, more laminar profile that appears "cleaner" rather than erroneous. The expert reads the setup text but cannot readily translate dimensionless parameters into quantitative visual expectations—for instance, mentally computing the hydrodynamic entry length $L_e \approx 0.05\,\mathrm{Re}\,D$ and checking whether the velocity profile in the image is fully developed at the appropriate downstream location. Claude appears to perform this cross-referencing explicitly.

6.2. Model-Specific Capabilities

The three VLMs exhibit distinct evaluation profiles that reflect different levels of cross-referencing capability.

Claude Opus 4.6 is the strongest cross-referencer. Its $\mathrm{FN} = 0$ with setup text means every error is caught, but its 33.3% image-only accuracy—the worst among all evaluators—confirms near-total reliance on setup–image comparison. The 55.6 pp drop is diagnostic: without textual parameters to compare against, Claude defaults to labeling most cases as plausible. Its high trial consistency (86.7–90.0% setup; identical 33.3% image-only across all three trials) suggests a deterministic assessment strategy with minimal stochastic variation.

GPT-5.4 exhibits moderate cross-referencing capability (67.8% setup, -18.9 pp drop). Its notable trial variance (56.7–76.7% setup-conditioned) with monotonically increasing accuracy across trials (T1 < T2 < T3) is unusual and may reflect session-level adaptation effects within the API's default temperature sampling. GPT-5.4 misses more error types than Claude but retains more capability without setup text (48.9% vs. 33.3%).

Gemini 3.1 Pro shows the weakest setup dependency (-4.4 pp). Its setup-conditioned accuracy (57.8%) is lower than the expert's (73.8%), suggesting limited cross-referencing capability even when parameters are provided. However, its image-only accuracy (53.3%) is second only to the expert (66.7%), indicating some innate visual assessment capability that does not depend on textual context. Its high false-positive rate (5.3 per trial) suggests a conservative bias toward flagging anomalies.

6.3. Implications for CFD Automation

These findings yield four practical implications for deploying VLMs in CFD validation pipelines.

First, **setup text is mandatory**. Without problem parameters, no VLM matches the expert. Setup-conditioned prompting—including Reynolds number, boundary conditions, and expected flow regime—is essential, not optional, for achieving useful accuracy.

Second, **model selection matters**. Only Claude achieves $\mathrm{FN} = 0$ (zero missed errors). GPT-5.4 is a viable alternative but with lower accuracy and higher trial variance. Gemini's high false-positive rate limits its practical utility as a first-pass filter.

Third, **false positives are the trade-off**. Claude's perfect recall comes at the cost of 3–4 false positives per 30 items. A practical pipeline should use VLM evaluation to flag cases for expert review, not to auto-reject simulations.

Fourth, **API isolation is a protocol requirement**. The 10.7 pp inflation from API-isolated (88.9%) to subagent evaluation (99.6%) demonstrates that filesystem contamination is a concrete threat. Published VLM benchmarks should specify the isolation level used during evaluation.

6.4. Limitations

Several limitations should be considered when interpreting these results.

Sample size. The cross-model comparison uses 30 items, sufficient for a pilot benchmark establishing the rank-reversal phenomenon but limiting statistical power for fine-grained per-error-type analysis. Larger item sets would enable confidence intervals on per-error-type recall.

Single expert baseline. One domain expert in thermal-fluid CFD provided all annotations. The expert's 73.8% accuracy may not be representative of all CFD practitioners; multi-expert evaluation with inter-rater agreement would strengthen the comparison.

2D steady-state scope. All scenarios are two-dimensional RANS simulations. Three-dimensional visualizations, transient flows (LES/DNS snapshots), and multi-physics problems (conjugate heat transfer, multiphase) present additional challenges not addressed here.

Evaluation asymmetry. The expert evaluated 80 setup-conditioned items and 30 image-only items; VLMs evaluated 30 items in each condition. Direct comparison uses the common 30-item subset, but the expert's broader evaluation data are not fully utilized in the cross-model analysis.

Default temperature. All VLMs were evaluated at default API temperature settings. Setting temperature to zero might reduce trial variance, particularly for Gemini. However, default settings reflect practical deployment conditions and are therefore the more relevant test.

No open-source VLMs. Only commercial frontier models were tested. Open-source models such as LLaVA, Qwen2.5-VL, and InternVL may exhibit different performance patterns and are needed for reproducibility analyses.

Templated errors. The error taxonomy uses controlled single-parameter perturbations. Real-world CFD errors may involve multiple simultaneous issues, subtler parameter deviations, or entirely different error modes (e.g., diverged simulations, domain decomposition artifacts). The benchmark should be extended with naturally occurring errors from production pipelines.

Task design. Setup-conditioned evaluation inherently rewards text–image consistency checking more than deep physical understanding. We acknowledge this as a feature of the task design rather than a limitation per se, since the intended application—automated pipeline validation—naturally includes setup context alongside simulation output.

7. Conclusion

This paper introduced CFD-VisQA, the first benchmark for evaluating vision-language model capability in assessing physical plausibility of CFD flow field visualizations. The benchmark comprises 10 canonical thermal-fluid scenarios with 60 OpenFOAM cases, 258 flow field images, and 279 physics-grounded questions spanning 6 systematically generated error types. An API-isolated evaluation protocol with base64-encoded images was developed after discovering that subagent-based evaluation inflates accuracy from 88.9% to 99.6% due to potential filesystem contamination.

The central finding is a rank reversal between evaluation conditions. With setup text, Claude Opus 4.6 leads at 88.9% accuracy with zero false negatives, followed by the domain expert at 73.8%. Without setup text, Claude drops to 33.3%—the lowest among all evaluators—while the expert leads at 66.7% with only a 7.1 pp decline. This reversal reveals that VLM performance derives from systematic cross-referencing of stated parameters against visual features, not from visual understanding of flow physics. The expert's stability reflects gestalt assessment grounded in physical intuition, a capability no current VLM possesses.

For practical CFD automation, the results support a two-stage validation architecture: Claude with setup text serves as a first-pass automated validator with perfect recall (FN = 0), flagging suspicious cases for expert review to resolve false positives. The benchmark dataset, evaluation scripts, and VLM response transcripts are publicly available to support reproducible research on VLM capabilities in engineering simulation assessment.

Data Availability

The CFD-VisQA benchmark dataset, evaluation scripts, and VLM response transcripts are available at `https://github.com/yhjoo-kier/OpenFOAM` (directory: `papers/cfd_visual_qa/`).

Acknowledgements

This work was conducted under the framework of the research and development program of the Korea Institute of Energy Research (C6-2419-63).

Declaration of Competing Interests

The authors declare no competing interests.

References

[1] G. Calzolari, W. Liu, Deep learning to replace, improve, or aid CFD analysis in built environment applications: A review, Building and Environment 206 (2021) 108315. doi:10.1016/j.buildenv.2021.108315.

[2] C. Ding, K. P. Lam, Data-driven model for cross ventilation potential in high-density cities based on coupled CFD simulation and machine learning, Building and Environment 165 (2019) 106394. doi:10.1016/j.buildenv.2019.106394.

[3] Y. Joo, Automated indoor CFD analysis from a single 2D image via vision-language model abstraction, manuscript submitted for publication (2026).

[4] S. Pandey, R. Xu, W. Wang, X. Chu, OpenFOAMGPT: A retrieval-augmented large language model (LLM) agent for OpenFOAM-based computational fluid dynamics, Physics of Fluids 37 (3) (Mar. 2025). doi:10.1063/5.0257555.

[5] Z. Dong, Z. Lu, Y. Yang, Fine-tuning a large language model for automating computational fluid dynamics simulations, Theoretical and Applied Mechanics Letters 15 (3) (2025) 100594. doi:10.1016/j.taml.2025.100594.

[6] W. Chow, J. Shi, L. Li, J. Jia, PhysBench: Benchmarking and Enhancing Vision-Language Models for Physical World Understanding, in: International Conference on Learning Representations (ICLR), 2025, key: chow2025physbench.

[7] J. Burgess, J. J. Nirschl, L. Bravo-Sánchez, A. Lozano, S. R. Gupte, J. G. Galaz-Montoya, Y. Zhang, Y. Su, D. Bhowmik, Z. Coman, S. M. Hasan, A. Johannesson, W. D. Leineweber, M. G. Nair, R. Yarlagadda, C. Zuraski, W. Chiu, S. Cohen, J. N. Hansen, M. D. Leonetti, C. Liu, E. Lundberg, S. Yeung-Levy, MicroVQA: A Multimodal Reasoning Benchmark for Microscopy-Based Scientific Research, in: 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2025, pp. 19553–19564. doi:10.1109/cvpr52734.2025.01821.

[8] A. C. Doris, D. Grandi, R. Tomich, M. F. Alam, M. Ataei, H. Cheong, F. Ahmed, DesignQA: A Multimodal Benchmark for Evaluating Large Language Models' Understanding of Engineering Documentation, Journal of Computing and Information Science in Engineering 25 (2) (Dec. 2024). doi:10.1115/1.4067333.

[9] C. Picard, K. M. Edwards, A. C. Doris, B. Man, G. Giannone, M. F. Alam, F. Ahmed, From concept to manufacturing: Evaluating vision-language models for engineering design, Artificial Intelligence Review 58 (9) (Jul. 2025). doi:10.1007/s10462-025-11290-y.

[10] S. Pandey, A. Ottley, Benchmarking Vision-Language Models on Visualization Literacy, in: EuroVis 2025, 2025, key: pandeyottley2025visliteracy.

[11] H. Chen, M. Wang, Z. Bai, ClimateIQA: A Multimodal Benchmark for Climate-Related Visual Question AnsweringKey: chen2024climateIQA (2024).

[12] J. Ezemba, C. McComb, C. Tucker, Simulation vs. Hallucination: Assessing Vision-Language Model Question Answering Capabilities in Engineering Simulations, in: Proceedings of the 7th Workshop on Design Automation for CPS and IoT, ACM, 2025, pp. 1–9. doi:10.1145/3722573.3727826.

[13] X. Yue, Y. Ni, T. Zheng, K. Zhang, R. Liu, G. Zhang, S. Stevens, D. Jiang, W. Ren, Y. Sun, C. Wei, B. Yu, R. Yuan, R. Sun, M. Yin, B. Zheng, Z. Yang, Y. Liu, W. Huang, H. Sun, Y. Su, W. Chen, MMMU: A Massive Multi-Discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI, in: 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), IEEE, 2024, pp. 9556–9567. doi:10.1109/cvpr52733.2024.00913.

[14] J. Roberts, H.-C. Lu, S. Forbes, SciFIBench: Benchmarking Large Multimodal Models for Scientific Figure Interpretation, in: Advances in Neural Information Processing Systems (NeurIPS), 2024, key: roberts2024.

[15] P. Lu, H. Bansal, T. Xia, MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts, in: International Conference on Learning Representations (ICLR), 2024, key: lu2024mathvista.

[16] Z. Wang, M. Xia, L. He, CharXiv: Charting Gaps in Realistic Chart Understanding in Multimodal LLMs, in: Advances in Neural Information Processing Systems (NeurIPS), 2024, key: wang2024charxiv.

[17] N. Alampara, M. Schilling-Wilhelmi, M. Ríos-García, I. Mandal, P. Khetarpal, H. S. Grover, N. M. A. Krishnan, K. M. Jablonka, Probing the limitations of multimodal language models for chemistry and materials research, Nature Computational Science 5 (10) (2025) 952–961. doi:10.1038/s43588-025-00836-3.

[18] S. Pramanick, D. Suris, C. Vondrick, SPIQA: A Dataset for Multimodal Question Answering on Scientific Papers, in: Advances in Neural Information Processing Systems (NeurIPS), 2024, key: pramanick2024spiqa.

[19] L. Li, Y. Wang, R. Xu, P. Wang, X. Feng, L. Kong, Q. Liu, Multimodal ArXiv: A Dataset for Improving Scientific Comprehension of Large Vision-Language Models, in: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, 2024, pp. 14369–14387. doi:10.18653/v1/2024.acl-long.775.

[20] A. Somasekharan, D. Panigrahi, CFDLLMBench: A Benchmark for Evaluating Large Language Models on Computational Fluid DynamicsKey: somasekharan2025 (2025).

[21] L. Wang, L. Zhang, G. He, Evaluations of large language models in computational fluid dynamics: Leveraging, learning and creating knowledge, Theoretical and Applied Mechanics Letters 15 (3) (2025) 100597. doi:10.1016/j.taml.2025.100597.

[22] Y. Zhang, K. Zheng, F. Liu, Q. Zhang, Z. Wang, AutoTurb: Using large language models for automatic algebraic turbulence model discovery, Physics of Fluids 37 (1) (Jan. 2025). doi:10.1063/5.0247759.

[23] S. Futamura, T. Fukuda, Quantitative Evaluation and Domain Adaptation of Vision–Language Models for Mixed-Reality Interpretation of Indoor Environmental Computational Fluid Dynamics Visualizations, Technologies 14 (3) (2026) 157. doi:10.3390/technologies14030157.

[24] M. Elrefaie, F. Morar, A. Dai, F. Ahmed, DrivaerNet++: A Large-Scale Multimodal Car Dataset with Computational Fluid Dynamics Simulations and Deep Learning Benchmarks, in: Advances in Neural Information Processing Systems (NeurIPS), 2025, key: elrefaie2025drivaernet.

[25] T. Dai, F. Wang, Q. Chen, Application of large language model in the design of indoor air distribution systems for office environments, Building and Environment 285 (2025) 113647. `doi:10.1016/j.buildenv.2025.113647`.

[26] C. Alanis Ruiz, M. Loomans, T. van Hooff, A deep convolutional generative adversarial network (DCGAN) for the fast estimation of pollutant dispersion fields in indoor environments, Building and Environment 276 (2025) 112856. `doi:10.1016/j.buildenv.2025.112856`.

[27] A. Kashefi, A Misleading Gallery of Fluid Motion by Generative Artificial IntelligenceKey: kashefi2024 (2024).

[28] L. M. Schulze Buschoff, E. Akata, M. Bethge, E. Schulz, Visual cognition in multimodal large language models, Nature Machine Intelligence 7 (1) (2025) 96–106. `doi:10.1038/s42256-024-00963-y`.

[29] S. Jassim, A. Oussidi, X. Gao, GRASP: A Novel Benchmark for Evaluating Language Grounding and Situated Physics Understanding in Multimodal Language Models, in: International Joint Conference on Artificial Intelligence (IJCAI), 2024, key: jassim2024grasp.

[30] S. Ghaffari, N. Krishnaswamy, Exploring Failure Cases in Multimodal Reasoning About Physical Dynamics, Proceedings of the AAAI Symposium Series 3 (1) (2024) 105–114. `doi:10.1609/aaaiss.v3i1.31189`.

[31] C. Liu, R. Jiang, D. Wei, C. Yang, Y. Li, F. Wang, X. Yuan, Deep learning approaches in flow visualization, Advances in Aerodynamics 4 (1) (Apr. 2022). `doi:10.1186/s42774-022-00113-1`.

[32] C. Banerjee, K. Nguyen, C. Fookes, K. George, Physics-Informed Computer Vision: A Review and Perspectives, ACM Computing Surveys 57 (1) (2024) 1–38. `doi:10.1145/3689037`.